

Virtualizing the Stanley: Gaussian Splatting as Educational Gateway to Advanced CGI Techniques

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Figure 1: Documentation from the ‘Virtual Stanley’ project undertaken at the University of Iowa’s Stanley Museum of Art, featuring students Nate Ingalls, John Charlson, Calvin Czeschin, Fin Deevy, Gavin Egger and Lydia Meth.

Abstract

Recent technological advances have granted CGI enthusiasts unprecedented ease of access to stereophotogrammetric 3D scanning techniques using consumer-grade camera systems, low-cost processing software, and even smartphone applications. This technological democratization has enabled new methodologies for students to engage hands-on with high-level CGI concepts regardless of their level of previous instruction, giving rise to ‘gateway’ educational opportunities specifically using Gaussian Splatting and Stereophotogrammetry to help students explore and enact complex CGI techniques through easily implementable activities. One of these is my ‘Virtual Stanley’ activity, in which students use handheld cameras and mobile devices to research and 3D capture (via Gaussian splats) specific artworks and their surroundings on display within the extensive art collection at the University of Iowa’s Stanley Museum of Art. In addition to serving as excellent documentation for students to reference outside of the museum setting, these 3D datasets are able to be conceptually redeployed within an array of AR/VR, Virtual Production, VFX, Previsualization, 3D spatial design, and other CGI pipelines. During these activities, a variety of themes emerged, especially regarding the manner in

which these technologies enable expanded public and research access, support documentary and narrative storytelling, and serve as advanced visual/data records and preservation references for these inherently fragile/ephemeral cultural artifacts. Key to this specific activity has been exploring these art encounters in conversation with the work of 20th century philosopher Walter Benjamin, especially the ideas espoused within his 1935 essay ‘The Work of Art in the Age of Mechanical Reproduction.’ His theories on the nature of an art object’s intrinsic ‘Aura’ and its unique ‘Time/Space Presence’ raise a multitude of compelling questions during these technologically evolved mechanical reproduction engagements.

CCS Concepts

• **Computing methodologies** → Animation; Rendering; Image manipulation; Graphics systems and interfaces; **Virtual reality**.

Keywords

Virtual Production, AR/VR, Gaussian Splats, Realtime Rendering

ACM Reference Format:

C K Olsen. 2025. Virtualizing the Stanley: Gaussian Splatting as Educational Gateway to Advanced CGI Techniques. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Educators Forum (SIGGRAPH Educators Forum ’25)*, August 10–14, 2025, Vancouver, BC, Canada. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3721242.3734015>

1 Overview

Both cinematography and CGI utilize core concepts of perspective, relationality, parallax, lighting, and form to achieve the illusion of dimension. Building upon this base, Gaussian Splatting has proven to be a valuable technique when teaching introductory computer graphic concepts, especially for live-action focused film students. Importantly, Gaussian Splatting has become widely accessible via

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SIGGRAPH Educators Forum ’25, Vancouver, BC, Canada

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ACM ISBN 979-8-4007-1544-0/25/08

<https://doi.org/10.1145/3721242.3734015>

free / low-cost software and smart phone apps from such developers as Epic, Polycam, and Niantic enabling educators to easily deploy collaborative engagements for students to explore AR/VR, VFX, and other CGI related techniques using familiar handheld capture devices and common cinematic cameras. Initial student experiments inevitably revolve around their immediate surroundings, resulting in playful experiments (and many ‘dimensional selfies’), that quickly teach students the strengths, limitations, and quirks that accompany GS capture techniques. To encourage deeper explorations, I introduced my students to an extensive collection of capture subjects in the form of artworks and exhibits housed within the Stanley Museum of Art at the University of Iowa. These exhibits could be repeatedly scanned using different cameras, software, and capture approaches with almost no variability in position, lighting, or spacing, allowing students to study the works, posit and test capture methodologies, gather results, and refine their capture skills while simultaneously growing a collaborative asset library of 3D scanned datasets. These engagements enabled critical discourse surrounding technical concerns (such as data cloud integrity, mesh complexity, coverage mapping, and parallax design), as well as artistic inquiries surrounding Walter Benjamin’s ideas expressed in his ‘The Work of Art in Mechanical Reproduction’. Queries have included discussions on whether the dimensional interactivity and lighting-related peculiarities of Gaussian Splatting might actually convey certain aspects of an art object’s ‘aura’ in a way that older analog capture techniques could not, as well as whether digital sharing of art/sculpture/likeness through these techniques enables a truly democratized audience engagement experience. Our collective research has inspired many offshoot student projects, including augmented reality interactions with scanned digital avatars, and VFX endeavors utilizing student-scanned assets and environments. Most engaging has been collaborative student efforts to refine these artwork-focused scanning methodologies into interactive, and potentially publicly navigable, ‘Virtual Stanley’ Museum exhibits. Ultimately, despite their simplicity, these Gaussian Splatting activities have proven to be invaluable introductions to essential computer graphics theory. Through these activities, students have unlocked critical thinking skills regarding the translational power of CGI, the nature of an object’s ‘Aura’, and more importantly, the nature of ephemerality and spatiality within their artistic works.

2 Metadata

This range of activities is suited for students of any level, especially cinematic arts students new to CGI techniques. These activities reference the following ACM CCS Computer Graphics subtopics:

- | | |
|----------------------------|------------------------|
| (1) Animation | (5) Image Manipulation |
| (2) Fundamentals | (6) Rendering |
| (3) Graphics I& Interfaces | (7) Visualization |
| (a) Mixed/Aug. Reality | |
| (b) Perception | |
| (c) Virtual Reality | |

Table 1: EETA metadata in tabular format.

Summary	This activity encourages students to explore Gaussian Splatting as an entrance point to advanced computer graphics techniques to correlate real and virtual camera processes, and their ability to generate virtual representations of real-world subjects..
Learning Outcomes	Students will examine relationships between Camera/Subject locality, parallax and data-point mapping; and output of assets for AR/VR, VFX and advanced CGI Pipelines.
Classification(s)	Animation; Fundamentals; Mixed/Aug. Reality; Perception; Virtual Reality; Image Manipulation; Rendering; Visualization
Audience	This exercise is ideal for student filmmakers of all levels interested in learning CGI techniques.
Dependencies	Students should have intermediate experience using real world photographic and/or cinematic production techniques.
Prerequisites	Introduction to photography and/or film/video production
Strengths	This exercise engenders collaborative work, research and development practices, and provides students with multiple opportunities for individual research, poster development, and portfolio exemplars.
Weaknesses	Advanced/custom re-deployment of student captured 3D assets for VFX, Gaming and AR/VR may still require high-end hardware and software.
Variants	Activity is not subject dependent and can be adapted for local points of interest/accessibility.
Assessment	This activity is not directly assessed, but capture processes and results are easily evaluated for quality and usefulness within production pipelines for final projects.

3 Materials

Gaussian Splatting benefits from high-end camera and lens systems, but suitable educational results can be achieved using entry-level capture devices, including HD/SLR cameras, smartphones and tablets. Processing requires a quality Gaussian Splat focused application, but a variety of open-source and low-cost options are readily available. Additionally, once processed, these GS datasets are importable into a variety of common graphics programs for further student exploration.

Acknowledgments

The author thanks the staff and leadership at the University of Iowa and Stanley Museum of Art, the SIGGRAPH Education Committee, and his Alternative Forms Fall '24 student cohort and Spring '25 undergraduate research team.